

Computer Programming Assignment (CS002)

This assignment contains solutions for all tasks related to Variables, Statements, Loops, and Functions in C Programming.

Task 1 - Q1: Temperature Conversion

```
#include <stdio.h>

int main()
{
    float celsius, fahrenheit;

    printf("Enter temperature in Celsius: ");
    scanf("%f", &celsius);

    fahrenheit = (celsius * 9 / 5) + 32;

    printf("Temperature in Fahrenheit = %.2f", fahrenheit);

    return 0;
}
```

Task 1 - Q2: Swap Two Numbers

```
#include <stdio.h>

int main()
{
    int num1, num2;

    printf("Enter first number: ");
    scanf("%d", &num1);

    printf("Enter second number: ");
    scanf("%d", &num2);

    printf("\nBefore Swapping\n");
    printf("First Number = %d\n", num1);
    printf("Second Number = %d\n", num2);

    num1 = num1 + num2;
    num2 = num1 - num2;
```

```
    num1 = num1 - num2;

    printf("\nAfter Swapping\n");
    printf("First Number = %d\n", num1);
    printf("Second Number = %d\n", num2);

    return 0;
}
```

Task 2 - Q3: Leap Year Program

```
#include <stdio.h>

int main()
{
    int year;

    printf("Enter a year: ");
    scanf("%d", &year);

    if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0))
    {
        printf("%d is a Leap Year", year);
    }
    else
    {
        printf("%d is not a Leap Year", year);
    }

    return 0;
}
```

Task 2 - Q4: Simple Calculator

```
#include <stdio.h>

int main()
{
    float num1, num2, result;
    char operator;

    printf("Enter first number: ");
    scanf("%f", &num1);

    printf("Enter operator (+, -, *, /): ");
    scanf(" %c", &operator);
```

```

printf("Enter second number: ");
scanf("%f", &num2);

switch(operator)
{
    case '+':
        result = num1 + num2;
        printf("Result = %.2f", result);
        break;

    case '-':
        result = num1 - num2;
        printf("Result = %.2f", result);
        break;

    case '*':
        result = num1 * num2;
        printf("Result = %.2f", result);
        break;

    case '/':
        if(num2 == 0)
        {
            printf("Division by zero is not allowed");
        }
        else
        {
            result = num1 / num2;
            printf("Result = %.2f", result);
        }
        break;

    default:
        printf("Invalid Operator");
}

return 0;
}

```

Task 3 - Q5: Prime Number

```

#include <stdio.h>

int main()
{
    int number, i, prime = 1;
    printf("Enter a number: ");
}

```

```

scanf("%d", &number);

if(number <= 1)
{
    prime = 0;
}

for(i = 2; i < number; i++)
{
    if(number % i == 0)
    {
        prime = 0;
        break;
    }
}

if(prime == 1)
{
    printf("%d is a Prime Number", number);
}
else
{
    printf("%d is not a Prime Number", number);
}

return 0;
}

```

Task 3 - Q6: Fibonacci Series

```

#include <stdio.h>

int main()
{
    int n, first = 0, second = 1, next, i;

    printf("Enter number of terms: ");
    scanf("%d", &n);

    printf("Fibonacci Series:\n");

    for(i = 1; i <= n; i++)
    {
        printf("%d ", first);

        next = first + second;
        first = second;
    }
}

```

```
        second = next;
    }

    return 0;
}
```

Task 3 - Q7: Reverse Number

```
#include <stdio.h>

int main()
{
    int number, reverse = 0, remainder;

    printf("Enter a number: ");
    scanf("%d", &number);

    while(number != 0)
    {
        remainder = number % 10;
        reverse = reverse * 10 + remainder;
        number = number / 10;
    }

    printf("Reversed Number = %d", reverse);

    return 0;
}
```

Task 4 - Q8: Circle Functions

```
#include <stdio.h>

float circleArea(float r)
{
    return 3.1416 * r * r;
}

float circleCircumference(float r)
{
    return 2 * 3.1416 * r;
}

int main()
{
    float radius, area, circumference;
```

```

printf("Enter radius: ");
scanf("%f", &radius);

area = circleArea(radius);
circumference = circleCircumference(radius);

printf("Area of Circle = %.2f\n", area);
printf("Circumference of Circle = %.2f", circumference);

return 0;
}

```

Task 4 - Q9: Palindrome Number

```

#include <stdio.h>

int isPalindrome(int n)
{
    int original, reverse = 0, remainder;

    original = n;

    while(n != 0)
    {
        remainder = n % 10;
        reverse = reverse * 10 + remainder;
        n = n / 10;
    }

    if(original == reverse)
    {
        return 1;
    }
    else
    {
        return 0;
    }
}

int main()
{
    int number;

    printf("Enter a number: ");
    scanf("%d", &number);

    if(isPalindrome(number))

```

```
{
    printf("%d is a Palindrome Number", number);
}
else
{
    printf("%d is not a Palindrome Number", number);
}

return 0;
}
```
